Universal counting: exact constructions and remaining obstructions

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Abstract

The target is exact counting for arbitrary Boolean circuits in time polynomial in their description length. This target is not achieved here. We construct a polynomial-size circuit-to-cubic compiler without expanding the output polynomial. Constant nonsingular Gaussian fibers and a two-value alphabet can retain the entire counting problem. A sharper affine matrix-pencil construction locates the same problem one nilpotency level beyond a verified polynomial case: with constant sources, an affine pencil with $A^3 = 0$ reduces to a quadratic count, whereas $A^4 = 0$ already represents every Boolean cubic gap. Conditions, normalization factors, original input counts and arithmetic costs are explicit. Additional attacks examine recursive replacement summaries and finite-field packing. Positive repairs acquire cubic and quartic etale norm geometry from raw coefficients, integrate coupled Gaussian radicals, and count verified field-linear programs at unbounded nilpotency depth. A Hessian Pfaffian manufactures norm coefficients erased by the absolute trace, with a fixed-degree reconstruction principle beyond quartics. None of the architecture-specific obstructions is a lower bound for arbitrary counting algorithms. The previous acquired chain-algebra theorem and its source files are preserved.

1 The target and what has not been proved

Given a Boolean circuit C with n declared independent inputs and description length L, compute

$$\#C = |\{x \in \mathbb{F}_2^n : C(x) = 1\}|$$

in $L^{O(1)}$ deterministic bit operations. The description includes its input inventory, so $n \le L$; otherwise merely writing the answer could exceed the bound. An output count has at most $n + 1$ bits. This target would establish polynomial computation of all #P functions, and decide SAT in polynomial time by testing positivity. Neither conclusion follows from the fixed-width subexponential theorem in the governing manuscript [1].

This work does not supply that polynomial counter. It attempts concrete extensions of the acquired representation and identifies where exact computation still contains the target. A reduction into a representation is not a counting algorithm. A proof that a specified summary loses information is not an impossibility theorem for another representation.

2 A direct circuit compiler with polynomial description size

Allow gates of fan-in at most two, unrestricted fan-out, and topologically ordered references. Each gate has a degree-two Boolean ANF with at most four monomials. For g internal gates, introduce wire variables y_j, their quadratic equations $q_j(x, y) = 0$, and the accepted-output equation $1 + o(x, y) = 0$. Introduce one multiplier per equation and construct

$$P_C(x, y, \lambda) = \sum_{j=1}^{g} \lambda_j q_j(x, y) + \lambda_{g+1}(1 + o(x, y)).$$

Every gate is handled only in its named wires. Its output is never expanded as a polynomial of the original inputs.

Proposition 2.1 (Exact circuit normalization)**.** *P_C is a cubic on $N = n + 2g + 1$ bits, with at most $5g + 2$ monomials, and*

$$\operatorname{gap}(P_C) := \sum_{x,y,\lambda} (-1)^{P_C(x,y,\lambda)} = 2^{g+1} \#C.$$

It is constructed in polynomial bit time and space, without an output ANF-size hypothesis.

Proof. For fixed x, y, a multiplier sum is two when its equation vanishes and zero otherwise. Their product selects exactly the valid accepting computations. Internal wires are uniquely determined by the original input, so the only excess factor is 2^{g+1}. Each gate equation has at most five monomials; its fresh multiplier raises degree by one. □

Index-list output uses $O((n + g)\log(n + g + 2))$ bits. The executable uses N-bit masks, taking $O((n + g)^2)$ bits. Charging even worst-case set operations, $O(L + (n + g)^3)$ bit time is conservative. Reconstruction divides an $O(n + g)$-bit integer by the known power 2^{g+1}.

This is standard gate arithmetization, not a claimed breakthrough or an improvement of the circuit-to-CNF corollary. It supplies a direct executable source for the stronger target. Thirty-six circuits passed independent input evaluation and full enumeration of their cubic phases, covering 108744 phase assignments. A 128-input OR circuit compiles into 383 phase variables and 510 monomials despite having $2^{128} - 1$ output ANF terms. Its known count is not claimed as an execution of a new general algorithm.

3 Two Gaussian states can retain the entire source count

Write a Boolean cubic as $p(z) = q(z) + \sum_{j=1}^{s} a_j b_j c_j$, where q has degree at most two and each a_j, b_j, c_j is an input coordinate. Introduce one pair (u_j, v_j) per cubic monomial and another pair for the lower-degree remainder. The elementary identity

$$\sum_{u,v \in \mathbb{F}_2} (-1)^{uv + \alpha u + \beta v} = 2(-1)^{\alpha\beta}$$

gives a cubic phase

$$F(z, u, v) = \sum_{j=1}^{s} (u_j v_j + u_j a_j + v_j b_j c_j) + u_* v_* + u_* q(z) + v_*.$$

All auxiliary polar matrices are the same nonsingular sum of hyperbolic planes. There is no pure-selector phase or external weight. Nevertheless, with $r = s + 1$,

$$G(z) = \sum_{u,v} (-1)^F = 2^r (-1)^{p(z)}, \qquad \sum_z G(z)^k = \begin{cases} 2^{rk + \dim z}, & k \text{ even}, \\ 2^{rk} \operatorname{gap}(p), & k \text{ odd}. \end{cases}$$

Thus every even moment is uninformative about the count; every odd moment retains exactly the original problem. The two-state distribution exists and is verified, but manufacturing its multiplicities remains unresolved.

Global Gaussian elimination is a stronger comparison than independent block propagation. For a phase $c(z) + q_0(x) + \ell(z)^T x$ with constant nonsingular polar matrix B, acquire $t(z) = B^{-1}\ell(z)$. Completing the square gives $G(q_0)(-1)^{c(z) + q_0(t(z))}$. The implementation recovers the source phase after hidden invertible mixing of the Gaussian coordinates. Degree can increase: eliminating $xy + xab + ycd$ gives $2(-1)^{abcd}$. Reducing the variable count can therefore move complexity into the residual expression. The six-variable example has fully checked gap 28.

4 An exact nilpotency boundary for affine Gaussian pencils

Let $z \in \mathbb{F}_2^m$, let $A(z)$ be an explicitly given affine r by r matrix, and let $a, b \in \mathbb{F}_2^r$ be constant. Define

$$\Phi(z, u, v) = u^T(I + A(z))v + a^T u + b^T v.$$

When $I + A(z)$ is invertible, character orthogonality yields

$$\sum_{u,v} (-1)^{\Phi(z,u,v)} = 2^r (-1)^{b^T (I+A(z))^{-1} a}. \tag{1}$$

The auxiliary polar matrix has blocks $0, I + A, (I + A)^T, 0$, so it is nonsingular at every such parameter. The entire phase is cubic, all parameter dependence in its quadratic coefficients is affine, and its linear sources are constant.

Theorem 4.1 (Verified polynomial case)**.** *If $A(z)^3 = 0$ for every Boolean z, then the exact signed and zero counts of Φ are computable in polynomial bit time. The condition itself is checked by polynomial-size coefficient identities, without enumerating z.*

Proof. In the Boolean polynomial ring, verify all coefficients of A^3 vanish. They have degree at most three. Then $(I + A)^{-1} = I + A + A^2$, and $p(z) = b^T(I + A + A^2)a$ has degree at most two. Construct its coefficients and evaluate its binary Gaussian exactly. Equation (1) gives $\operatorname{gap}(\Phi) = 2^r \operatorname{gap}(p)$. The total independent dimension is $m + 2r$, so the zero count is $(2^{m+2r} + \operatorname{gap}(\Phi))/2$. □

With $B_j = \sum_{i=0}^{j} \binom{m}{i}$, ordinary sparse coefficient products use at most $O(r^3 B_{j-1}(m + 1))$ operations to form A^j, each on $O(m)$-bit masks. The nilpotency-three test therefore admits the conservative bit bound $\widetilde{O}(r^3(m + 1)^4)$ with ordered coefficient storage. For the literal hash-set implementation, even charging adversarial collisions, $O((m + r + 1)^{12})$ bit time and $O((m + r + 1)^6)$ bits of storage are conservative bounds for the nilpotency-three test and count. Gaussian elimination and reconstruction use $O(m + r)$-bit integers. This is a verified identity in Boolean ANF; a useful graph layering is not assumed as additional input.

Theorem 4.2 (Universality at the next nilpotency level)**.** *Every Boolean cubic p on m variables with M ANF monomials has a polynomially constructed affine matrix $A(z)$ of dimension $r = 2M + 2$, with $A^4 = 0$, and constant a, b, such that*

$$b^T(I + A)^{-1}a = p(z), \qquad \operatorname{gap}(\Phi) = 2^r \operatorname{gap}(p).$$

Hence this class already retains arbitrary exact circuit counting under the normalization of Section 2.

Proof. Make one directed path of length three per monomial, from a common source s to a common sink t, with two private internal vertices. Its three edge labels are the monomial's selector variables, padded with ones. Use A_{ij} for an edge from i to j. Every entry is affine, no path has length four, and $(A^3)_{st} = p$. There are no shorter paths from s to t, so $(I + A)^{-1}_{st} = p$. Take $a = e_t$, $b = e_s$. Summing u in (1) forces $v = (I + A)^{-1}e_t$, establishing the sign and index conventions exactly. □

The input and output descriptions are polynomial, and every conditional Gaussian has full rank and only the two values $\pm 2^r$. Thus regular fibers, two states, affine coefficient dependence, bounded inverse degree and fixed nilpotency index do not automatically supply an easier first moment. This is a path-sum construction with established branching-program ancestry; the elementary encoding is not asserted to be historically new.

Constant sources are necessary in the positive theorem. If $A_{01} = z_0$, $A_{12} = z_1$, and other entries vanish, then $A^3 = 0$. Replacing the constant sources by $z_2u_2 + z_3v_0$ gives residual $z_0z_1z_2z_3$, with fully enumerated ten-bit gap 112. An added arbitrary cubic selector phase would also invalidate the quadratic conclusion.

4.1 Executed repair and exact scope

The production program manufactures the residual ANF, finds its actual interaction components, and uses a Gaussian for each quadratic component. A larger-degree component is enumerated only within an explicit bit budget. The cost is polynomial preprocessing plus the sum of the recorded component enumerations; a bound on their size is not silently assumed.

Eighteen cubic sources passed 378 conditional Gaussian checks against the original phase. Three affine nilpotency-three instances passed without enumerating any selector assignments in the counter. Two of these were also checked by all selector fibers. A 214-bit nilpotency-four control whose residual is thirty independent cubic monomials was counted using 240 assignments: its residual gap is 6^{30} and its full gap is $2^{62}6^{30}$. Ordinary component factorization supplies this repair, so the example is not a practical advantage over an equally informed method. It demonstrates that nilpotency four does not make every instance hard. The complete circuit-to-pencil bridge also returned the correct count of three original accepted inputs.

An independent audit enumerated 926720 full assignments across four universal nilpotency-four pencils, four conjugated nilpotency-three pencils, and the varying-source counterexample. These counts are separate from the conditional Gaussian checks. No universal polynomial bound follows from any of these executions.

5 What specified linear-summary interfaces cannot retain

5.1 Low-degree moments can miss the required interaction

For r independent bits z_i, put

$$G_i(z_i) = \sum_{x_i,y_i} (-1)^{x_i y_i + x_i z_i + y_i} = 2(-1)^{z_i}, \quad w_c(z) = (-1)^{z_1+\cdots+z_r+c}.$$

Every signed proper-subset moment $\sum_z w_c(z) \prod_{i\in S} G_i(z_i)$ vanishes for both $c = 0, 1$. Because $G_i^2 = 4$, the same holds for every joint moment of total degree below r, after reducing its exponents. But the complete product sums to $(-1)^c 2^{2r}$. Those moment data alone therefore cannot reconstruct the answer. Merging the states into their product leaves only two values, but its first moment is then the target, as in Section 3.

This is a restriction on fixed low-order moments with access to the weight only through those moments. The full combined phase in this example is quadratic: global Gaussian elimination solves it in polynomial time, and that comparator was executed. The example does not exclude an acquired global change of representation.

5.2 Independent summaries with a fixed bilinear join

On r boundary bits z, a left quadratic phase can pin $z = a$:

$$\sum_t (-1)^{t\cdot(z+a)} = 2^r \mathbf{1}_{z=a}.$$

Let a right component supply $(-1)^{u\cdot z}$. Their joined answer is $2^r(-1)^{u\cdot a}$. The Walsh matrix $H_{u,a} = (-1)^{u\cdot a}$ satisfies $HH^T = 2^r I$, so its rank over the rationals or complexes is 2^r. If the two sides are summarized independently into dimension D and a fixed bilinear pairing must recover all joins, the answer matrix factors through dimension D, forcing $D \geq 2^r$. Summary acquisition on either side may itself be nonlinear; independence and fixed bilinear fusion are the actual restrictions.

Again global Gaussian elimination counts this family in polynomial time. The lower bound is not for nonlinear fusion, joint processing, adaptive coordinate changes or general tensor contraction. The governing replacement proof changes the representation before reconstruction and is not excluded.

5.3 The old hard critical measure does not remove the issue

For the previous $\mathbb{F}_8[t]/t^2$ construction, with N even, the acquired critical measure on the first N source bits is

$$\mu(a) = 2^{N-1}(1 + \mathbf{1}_{a=0}).$$

For every source cubic p and subset S,

$$\sum_a \mu(a)(-1)^{p(a)+\sum_{i\in S} a_i} = 2^{N-1}\left[\operatorname{gap}\left(p + \sum_{i\in S} a_i\right) + (-1)^{p(0)}\right].$$

The known exceptional-point contribution can be subtracted. The same low-order moment collision and full-rank Walsh interface then remain. This rules out an input-independent small linear summary supporting all future cubic weights in that architecture. It does not rule out a summary manufactured for one particular weight.

6 Why affine padding does not remove the bad residue field

Let T_p be the third Boolean polarization of a cubic phase. A binary affine change $x \mapsto Sx + a$ conjugates its full centroid by S; translation does not change the third form. Thus the isomorphism class of that algebra is intrinsic to this representation orbit.

Adding an independent quadratic phase contributes a zero third tensor. For a radical-free original tensor, the added coordinates form exactly the new radical, and quotienting the radical recovers the original tensor and its centroid. In particular, adding an unused bit to alter the total input dimension does not change an acquired residue field $\mathbb{F}_8$ into one of degree coprime to three.

More generally, if T and S are radical-free trilinear forms on disjoint spaces V, W, then

$$\operatorname{Cent}(T \oplus S) = \operatorname{Cent}(T) \times \operatorname{Cent}(S).$$

To see that cross blocks vanish, put $x \in V$ and $y, z \in W$ in the centroid identity. It gives $S(C_{WV}x, y, z) = 0$ for every y, z; radical freeness implies $C_{WV} = 0$. The other cross block vanishes similarly, and the diagonal conditions are exactly the two original centroid identities. Consequently, adjoining independent tractable cubic blocks and then applying an invertible affine disguise retains the bad primitive factor. Its residue field degree cannot be removed by relabeling the product algebra. A field extension containing $\mathbb{F}_8$ also has degree divisible by three.

These are obstructions only to affine changes, independent padding and literal field extension. They do not exclude coupled auxiliary encodings, nonlinear bijections, interpolation or a different counting algorithm. Such operations must preserve the input measure and the original observable as well as the displayed polynomial. The circuit and pencil constructions above do so through explicit normalization factors, but they retain the hard residual rather than eliminate it.

7 Testing a bootstrap of the governing replacement construction

Sections 3–4 of the pinned source manufacture replacement rows of degree R, use Q interpolation points, and reconstruct Taylor order $q = \lfloor dR/Q \rfloor$, where d is the product degree before replacement. The stated vector-examination bound contains

$$2Qr\left(\sum_{j\le \min(q,n)} 2^j \binom{n}{j}\right)(2D+1)^{2\tau n} 2^{2n/r}.$$

Reducing the local group size alone does not remove the Taylor or shared measurement costs. For the explicit auxiliary array $D = r^{2Q}$, polynomial array size implies $Q = O(\log n)$. When

$d = \Theta(n)$, the prescribed order remains $\Omega(n/\log n)$. A universal linear feature representation supporting evaluation of every degree-q multilinear polynomial has dimension at least $\sum_{j\le q}\binom{n}{j}$: evaluation at indicator vectors of subsets of size at most q gives the unitriangular inclusion matrix.

Appendix D already recursively contracts descendants and shares immutable descriptions. With size s, $\lambda = \lceil \log_2 s \rceil$, $M_j = O(s8^{-j})$ and $u_j = \Theta(\lambda 2^{-j})$, its mode enumeration has branch budget $s^{O(1)} \exp O(M_j/\sqrt{u_j})$. The logarithms sum to $O(s/\sqrt{\lambda}) + O(\lambda \log \lambda)$. Replacing only its leaves by the same counter does not remove the root aggregation term from this analysis. Packet-entry evaluation also has the charged cost $\exp O(L \log s)$. A short description is not itself a polynomial-time evaluation guarantee. These are limits of that bootstrap argument, not unconditional time lower bounds.

7.1 Source-level jet and precision failures

For negative 2-CNF on an even cycle, scale one bipartition by $1 - T$. At $T = 1 + z$ the summed polynomial is

$$P_n(z) = \operatorname{Tr} \begin{pmatrix} 1 & 1 \\ 1 & 1+z \end{pmatrix}^n.$$

For even $\ell \ge 4$, the matrix trace identity gives

$$P_{2\ell}(z) - P_\ell(z)^2 = -2z^\ell.$$

Thus the formulas on $C_{2\ell}$ and $C_\ell \sqcup C_\ell$ have the same first ℓ deformation coefficients, width two, and occurrence two, but their counts differ by two. This refutes that single-node truncation shortcut. Recognizing the 2-regular scope graph and acquiring its cycle components supplies an exact two-state transfer repair. The identity was checked at six lengths; 139776 assignments also independently verified the smaller original formulas.

There is a separate all-jet arithmetic obstruction for an actual source packet. Put $L = 2^a$, $S = \{-L, \dots, L-1\}$, $C(T) = \prod_{s\in S}(T - s)$, $\alpha = 2L$, and $w(T) = C(T)/(C'(0)T)$. Then $w(0) = 1$. At an active nonzero node t with $i = v_2(t) < u \le a$, the minimum nonzero coefficient valuation of $w(t + \alpha z)$ is $a + 1 - i$. Factoring out the simple root at t gives a leading factor α/t times a unit and higher-valuation corrections. Therefore every coefficient of $w(t + \alpha z)^m$ is divisible by $2^{m(a-u+2)}$ at every active node, although $w(0)^m = 1$. At that precision, retaining every derivative still loses the target bit. The companion proof supplies the exact derivative valuations and includes the source's root parameter relation $a = 2u$.

7.2 An actual packet interface retains arbitrary counting

An executed integer packet channel is

$$K = \begin{pmatrix} 1 & 6035 \\ 0 & -5964 \end{pmatrix}.$$

Its nonunit Smith entry has two-adic valuation two. Tensor powers therefore have invariant entries with valuations $2|S|$. The minimum universal separable width modulo 2^p is

$$\sum_{j=0}^{\min(m,\lfloor (p-1)/2 \rfloor)} \binom{m}{j},$$

which reaches 2^m at $p \ge 2m + 1$. This is a universal linear-interface bound, not a prohibition on input-dependent nonlinear aggregation.

The retained computation is explicit. For arbitrary Boolean circuit output $A(Y)$, set $B(X) = (-11999)^{m-|X|}$. Directly,

$$K \begin{pmatrix} -11999 \\ 1 \end{pmatrix} = -5964 \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Hence contraction of A, B and $K^{\otimes m}$ equals $(-5964)^m \sum_Y A(Y)$. Precision $3m+1$ suffices to recover the count after dividing 2^{2m} and inverting the remaining odd unit. Five small CNF instances execute this reduction. The compact kernel does not supply the missing aggregation algorithm.

For raw cubic boundary ANFs, full permutation invariance is instead recognized by checking each complete degree layer. The resulting phase depends only on Hamming weight modulo four. A manufactured transfer on the two weight residues has sixteen states. A nonlinear 64-variable-per-side test agrees with independent binomial aggregation. This is a restricted, classical symmetry repair with acquired conditions, not a general theorem.

8 Acquired cubic blocks and exact Gaussian restriction

8.1 Hidden independent triples

For a raw Boolean cubic on $n = 3r$ bits, suppose an unknown binary affine bijection turns it into a sum of independent three-variable cubics, each with nonzero cubic coefficient and arbitrary local quadratic and affine terms. Its third tensor is a direct sum of volume forms, so its full centroid is $\mathbb{F}_2^r$. Solving the centroid equations and splitting its primitive idempotents manufactures the blocks. The earlier product-algebra acquisition already supports these components; the new backend supplies their missing eight-point counts.

The compiler contracts the raw third tensor with the acquired basis to verify every cubic coefficient. After cancellation, values at zero, unit vectors and pairs reconstruct the complete quadratic remainder. It rejects any quadratic term crossing two components, assigns the constant exactly once, and multiplies the r local gaps. Thus all lower terms are retained, and the cost is polynomial in the explicit input plus $8r$ local evaluations. The construction does not receive a partition or planted coordinates.

Raw, densely disguised examples on 30 and 60 bits acquired 10 and 20 blocks. Their gaps were -27648 and 84934656, using 80 and 160 local assignments. Independent complete coefficient identities checked respectively 4526 and 36051 possible ANF coefficients. Adding one genuine cross-block quadratic changed the first answer to -9216; the independent-block gate correctly rejected it.

Composing this backend with the nilpotent Gaussian compiler counted an explicit 9078-bit phase. Its affine matrix had dimension 4524 and 6783 nonzero entries. Verified elimination manufactured the 30-bit residual; centroid acquisition then manufactured its ten blocks. The exact full gap was $-27648\, 2^{4524}$. This execution establishes a correct composition, not an advantage over a method supplied with the planted decomposition.

8.2 A reusable exact preprocessor: eliminate the third radical

Independent padding leaves the old centroid intact, but *coupled* Gaussian variables can impose restrictions that change the surviving cubic. The following construction handles that operation explicitly.

Theorem 8.1 (Acquired Gaussian restriction)**.** *Given any explicit Boolean cubic p on n bits, a deterministic polynomial algorithm returns either* $\operatorname{gap}(p) = 0$*, or an integer $e \geq 0$ and an explicit cubic p_* on $m \leq n$ bits with zero third-tensor radical, such that*

$$\operatorname{gap}(p) = 2^e \operatorname{gap}(p_*).$$

The terminal may instead be a constant on zero bits. No counting algorithm for the arbitrary radical-free terminal is assumed. Each nontrivial step strictly decreases the number of surviving bits.

Proof. Compute $R = \mathrm{rad}(T_p)$ by binary linear algebra and extend its basis to $V = U \oplus R$. Reconstruct the complete polynomial in these coordinates. Every term involving a coordinate of R has degree at most two, since its third coefficient vanishes identically.

If two radical coordinates have a quadratic coupling, write the phase as $xy + xL + yM + N$, where L, M are affine in the remaining bits. Exact summation over x, y gives $2(-1)^{N+LM}$. The Boolean product uses $z_i^2 = z_i$; constants and all newly created quadratic terms are retained. Continue until the remaining radical coordinates have no pairwise couplings. If $k = \dim R$ and t pairs were removed, there are $k - 2t$ such coordinates.

Each remaining coordinate w contributes $wL_w(u)$, hence its sum is $2\mathbf{1}_{L_w(u)=0}$. Solve these affine equations exactly. Inconsistency gives zero. Otherwise manufacture their solution space $u = u_0 + Bz$, of dimension $\dim U - \mathrm{rank}(L_w)$. The exact result is

$$\mathrm{gap}(p) = 2^{k-t}\,\mathrm{gap}\big(F(u_0 + Bz)\big),$$

where F includes all Gaussian corrections. There is no additional constraint-rank multiplier: B parametrizes each surviving u exactly once. The restricted cubic can have a new third radical, so acquire and eliminate it again. At most n dimension-decreasing steps occur. □

The implementation uses low-weight evaluations to recover all coefficients; it does not enumerate the full domain. A conservative charged bound is $O(L + n^8)$ bit operations and $O(L + n^5)$ bits of retained certificates, before any terminal counter, for the explicit bitmask input length L. Exact final counts have $O(n)$ bits. The executable `radical_gaussian_descent.py` exposes the terminal as an argument, allowing any independently certified acquired counter to be composed with the same normalization proof.

Fifty-five small random cases and a two-stage example checked 10363 original assignments. The two-stage example creates a new radical after its first restriction and has gap 48. An independent audit checked eleven targeted cases, 203 assignments and five complete affine solution sets. All passed, including inconsistent constraints and zero-dimensional terminals.

Two larger raw affine disguises also passed. A 46-bit coupled Gaussian extension reduced to the 30-bit hidden-block phase, with gap -7077888. A 36-bit example included arbitrary cubic terms involving three extra coordinates; acquired Fourier constraints forced those coordinates to zero, revealing the hidden 30-bit blocks and giving gap -221184. The raw coefficients and independent planted identities are saved. These examples show a useful acquired change of representation, while the arbitrary radical-free case remains unresolved.

9 Why unrestricted quadratic glue still contains arbitrary counting

Consider the sharply specified class

$$f = \sum_{i=1}^{r} x_{3i-2}x_{3i-1}x_{3i} + Q(x), \qquad \deg Q \le 2.$$

Its third tensor has no radical and has full centroid $\mathbb{F}_2^r$, regardless of Q. Nevertheless, there is a one-query polynomial metric reduction from arbitrary cubic-gap counting to zero counting in this class.

First note the exact invariant $v_2(\mathrm{gap}(f)) = r$. Expand $(-1)^f$ as the integer product of $(1 - 2m)$ over its nonconstant ANF monomials. A chosen set of k monomials covering u variables

contributes valuation $k + 3r - u$. The unique selection of valuation r is the set of all r disjoint cubic terms: every other selection has valuation at least $r+1$. Thus the gap is never zero. A pure nonzero scaling reduction from all source gaps is impossible in this representation; the reduction below uses modular reconstruction instead.

Let the source cubic p have N variables and M cubic monomials. Clone each cubic occurrence into a private triple and enforce equality to the original coordinates with three Fourier multiplier bits per triple. The resulting core F has $N_0 = N + 6M$ bits and

$$\operatorname{gap}(F) = 2^{3M} \operatorname{gap}(p).$$

Its M cubic triples are disjoint, but $K = N + 3M$ bits lie outside them. For each such bit x, add an anchor xab and t leaf triples $y_j c_j d_j$, coupled only by the quadratic terms xy_j. Summing the new bits gives the exact two-entry weight

$$(g_t(0), g_t(1)) = (4 \cdot 6^t, 2 \cdot 2^t) = 2^{t+1}(R, 1), \qquad R = 2 \cdot 3^t.$$

These weights are manufactured by explicit Boolean gadgets.

The padded target T has $r = M + K(t+1)$ disjoint triples covering every bit. For an integer polynomial P used only in the proof,

$$\operatorname{gap}(T) = 2^{K(t+1)} P(R), \qquad P(1) = \operatorname{gap}(F), \qquad |P(1)| \le 2^{N_0}.$$

Choose $t = N_0$. Then $R - 1 > 2^{N_0+1}$ and $R \equiv 1 \pmod{R-1}$, so

$$\operatorname{gap}(p) = 2^{-3M} \operatorname{centered_mod}\left(2^{-K(t+1)} \operatorname{gap}(T),\, 2 \cdot 3^t - 1\right).$$

The centered residue is exact for positive, negative and zero source gaps. From one zero-count answer obtain $\operatorname{gap}(T) = 2\#\{T = 0\} - 2^{3r}$. The target has $O((N+M)^2)$ bits, polynomially many monomials, and its count has $O((N+M)^2)$ output bits. The modulus has $O(N+M)$ bits. Construction and all divisions and residues have polynomial bit cost. Composing the circuit normalization in Section 2 establishes #P-completeness under these metric reductions. It is not a time lower bound.

The executable constructor evaluates no source assignments. Its tests checked 3108400 complete-domain assignments, with a separately labelled 197128 core assignments used only to evaluate large-target weighted oracles. Those large targets were not solved by a new general counter. Four additional circuit normalizations checked 480 phase assignments, including an unsatisfiable circuit. A dense 18-bit affine disguise acquired all six primitive blocks and retained ten genuine cross-block quadratic terms. The companion proof and program preserve the exact construction and readout.

10 Packing the original independent inputs into one field coordinate

Let C have n original inputs and g gates. Construct a binary field $K = \mathbb{F}_{2^m}$ with $n \le m < 2n$ by an explicit quadratic tower, choosing m as the next power of two. Trace-one tower elements supply each irreducible quadratic extension. Linear dependence among powers of the last generator manufactures a polynomial-basis presentation. This construction and the following trace-dual basis computation have polynomial bit cost.

For a basis e_i, invert $G_{ij} = \operatorname{Tr}(e_i e_j)$ to obtain its trace-dual basis β_i. The polynomials

$$L_i(X) = \operatorname{Tr}(\beta_i X) = \sum_{j=0}^{m-1} \beta_i^{2^j} X^{2^j}$$

return the original binary coordinates. Share the m Frobenius powers, then replace AND, XOR, NOT and OR by uv, $u+v$, $1+u$ and $u+v+uv$. Every gate remains binary-valued on K. The resulting arithmetic circuit $C_K(X)$ has $O(nm+g)$ nodes and has introduced no independent gate variables. Choose η with absolute trace one, and define

$$H(X) = \eta(1 + C_K(X)), \qquad Y^2 + Y = H(X).$$

Each accepted input has 2^{m-n} field preimages and each zero-trace right hand side has two Y solutions. Consequently

$$\#\{(X,Y) \in K^2 : Y^2 + Y = H(X)\} = 2^{m-n+1}\#C.$$

The description uses $O(nm^2 + (nm+g)\log(nm+g))$ bits; a conservative construction bound is $O(m^4 + g\log(nm+g))$ bit operations. The trace-one factor is essential: without it, even m would make a binary-valued right hand side have trace zero identically. An executed four-input example produced 32 points instead of the correct 16 when that factor was omitted.

This solves the independent-input bookkeeping problem but leaves a real counting problem. An arithmetic circuit does not supply its expanded polynomial at no cost. Expansion is explicitly budgeted in the program. Lauder–Wan's first-count bound for a two-variable ordinary degree-D equation in characteristic two is

$$\widetilde{O}(m^{13}D^6) \text{ time}, \qquad \widetilde{O}(m^8 D^4) \text{ space}.$$

These follow from Proposition 36 of [4]; they concern the first count, not merely writing the entire zeta function. A short arithmetic description or a small number of field coordinates does not remove D.

For pointwise character normalization over K, positive exponents reduce modulo $2^m - 1$, with a positive multiple represented by $2^m - 1$, never by zero. Doubling orbits combine coefficients with the appropriate Frobenius twist. Short orbits use relative traces, including their zero tests. These operations preserve the character on K; unlike a genuine Artin–Schreier coboundary, they need not preserve points over every extension.

A binary-degree-r phase can use exponents of binary weight at most r. Rotating a circular support pattern after its longest gap gives $D < 2^{m-\lceil m/r\rceil+1}$, and the norm exponent $1 + 2^a + \cdots + 2^{(r-1)a}$ with $m = ra$ shows the exponential order is sharp. The norm itself is nevertheless easy to sum by its fibers. Likewise, a 32-input NOR circuit produced 3156 arithmetic nodes but minimum normalized trace degree $2^{32} - 1$. Its count remains elementary; this is a failure of the low-degree shortcut, not a counting lower bound.

Thirty-two circuit packings and eighteen trace normalizations were verified on complete small fields. Gcd acquisition of a power-image subfield also passed dimensions $8 \to 2$, $32 \to 8$ and $64 \to 4$. If the nonzero image has constant fiber size g, its exact correction is $S_K = gS_F - (g-1)(-1)^{\mathrm{Tr}(c_0)}$. A noncanceling linear term makes that gcd one, which motivates the next repair.

11 A raw Boolean compiler for cubic norm geometry

Theorem 11.1 (Acquired cubic etale norms)**.** *Let $a \geq 3$, $F = \mathbb{F}_{2^a}$, and let A be any cubic etale F-algebra:*

$$A \cong F^3, \quad F \times \mathbb{F}_{2^{2a}}, \quad \text{or} \quad \mathbb{F}_{2^{3a}}.$$

For $C \neq 0$, $B \in A$, an arbitrary binary affine bijection $\phi : \mathbb{F}_2^{3a} \to A$, and $\epsilon \in \mathbb{F}_2$, consider

$$p(x) = \mathrm{Tr}_{F/\mathbb{F}_2}\left(CN_{A/F}(\phi(x)) + \mathrm{Tr}_{A/F}(B\phi(x))\right) + \epsilon.$$

From only its raw Boolean ANF and $n = 3a$, the compiler manufactures the field, the cubic algebra, norm coordinates, affine center and linear element, verifies the complete pointwise identity, and counts exactly. Conservative costs for input bit length L are

$$O(n^{10} + Ln^3) \text{ bit time}, \qquad O(n^6 + L) \text{ bits of space}.$$

It introduces no new independent inputs and no counting multiplicity.

Proof. Over a splitting field, the norm is a product of three independent linear forms. Its third tensor is a traced F-volume form. Its full balanced binary centroid is exactly F: putting arguments in repeated coordinate lines forces every off-diagonal block of a centroid operator to vanish, and the remaining trace identities force all diagonal maps to be multiplication by the same field element. The trace pairing is nondegenerate. Binary linear algebra and the retained field-acquisition routine therefore manufacture F and a three-dimensional free module, even after an arbitrary disguise.

In these coordinates, the third tensor fixes the coefficient $\mu \neq 0$ of $X_1X_2X_3$. Fit the remaining quadratic binary coefficients using the twelve field-parameter families X_i^3, $X_i^2X_j$ for $i \neq j$, and X_iX_j for $i < j$. Their $12a$ binary polar columns are independent when $a \geq 3$: cross-line relations involve the distinct linearized exponents $1, 2, 2^{a-1}$, while a single-line relation has polar $cx^2 + \sqrt{c}\, x^{2^{a-1}}$, which vanishes identically only for $c = 0$. This uniquely separates an ordinary homogeneous cubic H, its cross quadratic coefficients, and its binary affine remainder.

The norm algebra is then acquired from H, rather than supplied. Examine the seven points $(1, s, s^2)$ for distinct $s = 0, \ldots, 6$ in F. For a norm, each geometric linear factor restricts to a nonzero polynomial of degree at most two, so their product has at most six roots. One point is a direction v with $H(v) \neq 0$. With v fixed, the same seven-point search finds u for which $g(T) = H(u + Tv)$ is squarefree: the three pairwise root-coincidence conditions have product degree at most six and are nonzero. Normalize g to be monic and form $A' = F[r]/g(r)$.

Set $L_i = \partial_i H(u + rv) \in A'$. At each geometric root, the slice point lies on exactly one factor plane and its gradient is a nonzero multiple of that factor's normal. Thus $L(X) = \sum_i L_i X_i$ is invertible. The program verifies this determinant and every coefficient of $H(X) = C' N_{A'/F}(L(X))$. There are only ten possible homogeneous coefficients. This test rejects arbitrary cubics with the same third tensor but different geometry. Computing $\gcd(g, T^q - T)$ acquires the field-factor pattern without enumerating F.

Writing the cross quadratic coefficients as P_{ij}, the unique center is $\mu^{-1}(P_{23}, P_{13}, P_{12})$. Only the XYZ term creates cross quadratic terms under translation; the other translated repeated-index cubics produce squares, which become binary linear terms under the trace. Check every remaining binary quadratic coefficient to be zero. The nondegenerate etale trace pairing then recovers the remaining element B'. Finally reconstruct and compare every raw ANF coefficient from the manufactured norm expression. The verified expression has degree at most three, so this full coefficient check requires only $\sum_{j=0}^{3} \binom{n}{j}$ low-weight probes.

All steps are explicit linear algebra, constant-degree polynomial arithmetic, and the exact Kloosterman evaluation below. Charging the field, matrix, coefficient and integer operations gives the stated conservative bounds. □

11.1 Evaluation and all nonunit cases

Let $K_F(t) = \sum_{y \neq 0} (-1)^{\mathrm{Tr}(y + t/y)}$. If B' is a unit and A' has s field factors, Davenport–Hasse lifting and multiplicative Fourier inversion give

$$S = (-1)^{3-s} q K_F(N(B')/C'), \qquad q = 2^a.$$

This exponential-sum identity has established ancestry [5]; the contribution here is the acquired and verified route to its inputs. For nonunits the remaining sum is explicit:

A'	B'	S
$\mathbb{F}_{q^3}$	0	$-q^2 - q$
$F \times \mathbb{F}_{q^2}$	$(0, d)$	q
$F \times \mathbb{F}_{q^2}$	$(b, 0),\ b \neq 0$	$q(q+1)$
F^3	all components zero	$2q^2 - q$
F^3	1 nonzero component	$q(q-1)$
F^3	2 nonzero components	$-q$

These follow by summing the linear factor first and counting the zero or nonzero norm fiber. Restore the acquired constant sign afterwards. The retained finite Dwork backend uses precision $a+2$, matrices of size $8(a+2)-1$, and a Frobenius factors. The elementary bound $|K_F(t)| \le q-1$ gives unique centered integer reconstruction; no field enumeration occurs in this backend.

For a fixed acquired base norm p_0, there is also an intrinsic admissibility test for a quadratic perturbation Q. Its polar must satisfy

$$B_Q(u,v) = T_{p_0}(t,u,v)$$

for some t. The contraction map has n columns and $n(n-1)/2$ rows and is injective because the third radical is zero. Thus it manufactures the unique translation exactly, allowing arbitrary linear remainder without assuming low quadratic rank. This test is relative to that fixed base norm; its rejection does not exclude a different norm geometry.

11.2 Executed evidence and composition

Fourteen raw affine-disguised norms passed, covering all three etale types and all zero/nonzero linear-component regimes. Thirteen complete original truth tables checked 275456 assignments. A 24-bit example with 1179 raw ANF terms had gap -768 and 8388224 zero outputs. Its independent oracle summed the planted Kloosterman expression over F_{256}, while the production algorithm acquired the algebra and used Dwork. The complete 24-bit raw identity was checked on 2325 coefficient-recovery points. A smooth non-norm cubic and an outside-lift quadratic were rejected.

An independent audit made the first six direction candidates fail on a valid F_8 split norm and verified success of the seventh. It also checked the polar-fit ranks $36, 48, 60$ in field degrees $3, 4, 5$, and the real alias at degree two, whose rank is only 15 instead of 24. Thirty targeted unit/nonunit checks used six independent Walsh tables and 3072 assignments.

Composing this raw norm compiler with acquired radical Gaussian descent counted a 13-bit example against all 8192 original assignments, and a 40-bit disguised Gaussian extension with exact gap -196608. The latter manufactured a 24-bit norm terminal and eight Gaussian factors; its independent source normalization and complete raw norm certificate agreed.

The earlier explicit field-support counter also handles $\mathrm{Tr}_K(cX^{q^2+q+1} + dX^{q+1} + bX + e)$ for $K = \mathbb{F}_{q^3}$. Writing $C = \mathrm{Tr}_{K/F}(c) \neq 0$, translation by $t = (d/C)^q$ removes the quadratic orbit and leaves linear coefficient $B = b + Ct^{q+q^2}$ and constant $e' = e + bt$. The norm–Kloosterman formula then applies. In its 48-bit execution, $\mathrm{Tr}(X^{4295032833} + X)$ had gap -29425664, with residual $K_{F_{65536}}(1) = -449$. An independent elliptic Frobenius recurrence verified that value. These support and raw-acquisition compilers are restricted counters; neither makes arbitrary packed circuits admissible.

12 Removing the fixed nilpotency bound in a field-linear description

The Boolean pencil obstruction does not mean that every depth-four residual is hard. Changing the coefficient domain gives a fully executed counter with *unbounded* matrix dimension and nilpotency index, for a restricted but explicit computational description.

Let $F = \mathbb{F}_{2^d}$ be given by an irreducible binary modulus, let $A(Z)$ be an explicit affine r-by-r matrix over $F[Z]$, and let $a, b \in F^r$ be constant vectors. These are the source description, not a supplied inverse, residual polynomial or hidden block decomposition. Verify $A(Z)^r = 0$ by exact polynomial matrix multiplication. Reject if the identity fails.

Theorem 12.1 (Verified field-pencil counter)**.** *On every such accepted input, the Boolean phase on $N = d(2r+1)$ original bits defined by*

$$\Phi(Z,u,v) = \mathrm{Tr}_{F/\mathbb{F}_2}\big(u^T(I + A(Z))v + a^T u + b^T v\big)$$

has an exact deterministic polynomial-time count, uniformly in d and r. The inverse and all arithmetic costs are constructed and charged. No fixed nilpotency bound, small matrix width, or field enumeration is required.

Proof. The polynomial identity gives $(I+A)^{-1} = I + A + \cdots + A^{r-1}$. For each Z, sum first over u to impose $(I+A)v = a$, which has exactly one solution. The u sum contributes q^r, where $q = 2^d$. Therefore

$$\operatorname{gap}(\Phi) = q^r \sum_{Z \in F} (-1)^{\operatorname{Tr} P(Z)}, \qquad P(Z) = b^T \sum_{j=0}^{r-1} A(Z)^j a, \quad \deg P \le r-1.$$

Manufacture every coefficient of P during the same matrix-power loop. This is a small ordinary polynomial because each matrix entry has ordinary degree at most one in a *single field coordinate*.

Apply the retained finite additive-Dwork construction to the explicit univariate P. Its torus sum is computed at precision $d+2$, with d twisted Frobenius factors and an interval of $O(rd)$ states. Add the zero point contribution $(-1)^{\operatorname{Tr} P(0)}$. The elementary absolute bound $q-1$ on the torus sum makes centered reconstruction unique. Finally multiply by 2^{dr}, and return $(2^N \pm \operatorname{gap}(\Phi))/2$ as the zero and one counts. □

With dense indexed polynomial matrices and schoolbook arithmetic, the matrix-power stage has $\widetilde{O}(r^5 d^2)$ bit cost: multiplication by the affine input matrix makes the kth step use $O(r^3 k)$ field operations. The finite-Dwork stage has conservative cost $\widetilde{O}(r^3 d^8)$ and storage $\widetilde{O}(r^2 d^4)$. Matrix coefficients add $\widetilde{O}(r^3 d)$ storage. A dense source description has $O(r^2 d + r^2 \log(r+2))$ bits; sparse indexing is also supported. If all dictionary operations in the literal Python prototype are charged pessimistically, $O((L+r+d)^{12})$ bit time remains a conservative polynomial implementation bound. Output integers have $O(dr)$ bits.

For clarity, the growing-degree truncation is explicit. Remove the constant character sign and set $D = \deg P > 0$, $\nu = d+2$, $M = 4D\nu$. In the unramified ring modulo 2^ν, use the standard splitting series $\theta(T) = \exp(-2T + 2T^2)$ as a formal coefficient series; its coefficients satisfy $v_2(\theta_j) \ge j/4$. The local Frobenius matrix has entries

$$A_{u,v} = [Z^{2u-v}] \prod_{e>0} \theta([c_e] Z^e), \qquad 0 \le u, v < M,$$

where $[c_e]$ denotes the manufactured Teichmuller lift. A coefficient of degree s has valuation at least $s/(4D)$. On a closed Frobenius walk, the sum of its nonnegative coefficient exponents $2u - v$ equals the sum of its states. Any omitted walk visiting a state at least M therefore vanishes modulo 2^ν. The standard additive-Dwork trace identity gives the torus sum as $(q-1)$ times the trace of the d twisted matrices. This justifies the polynomial state bound when D grows; it does not assume a small cohomology quotient. The formal splitting series must be evaluated through its coefficients, not by substituting into an ordinary exponential.

This is a construction from an explicit field-linear matrix program, using established Gaussian elimination and Dwork point counting. It is not an acquisition theorem for arbitrary raw Boolean circuits and is not claimed as a historically new point-counting principle.

The executable tested twelve small pencils, including arbitrary affine coefficients and constant sources, against ninety independent conditional binary-Gaussian sums and complete residual field sums. A larger input used $d = 16$, $r = 10$, and a dense binary similarity hiding two paths of lengths seven and three. The verified nilpotency index was eight, and the manufactured residual was $Z^7 + Z^3$. Its Dwork calculation used precision 18, 504 states and sixteen Frobenius factors, with no production field scan. An independent sum over all 65536 field points gave -1024. Thus the 336-bit original phase has

$$\operatorname{gap}(\Phi) = -2^{170}, \qquad \#\{\Phi = 0\} = 2^{335} - 2^{169}, \qquad \#\{\Phi = 1\} = 2^{335} + 2^{169}.$$

The independent field scan is labelled as an oracle. These observations do not establish a comparative practical speed advantage.

The same construction directly counts the state program $Z \mapsto \operatorname{Tr}(b^T(I + A(Z))^{-1}a)$ on its original d independent input bits. Its r field-valued internal states are uniquely determined by Z; they are not new independent inputs. The manufactured P is its exact output polynomial. In the executed example, this 16-input-bit program has 32256 zero outputs and 33280 one outputs. The larger 336-bit phase above is a separate exact Gaussian realization with the displayed normalization.

12.1 Why arbitrary Boolean selector packing does not inherit the bound

The universal Boolean pencil uses many independent binary selectors in its entries. Packing those selectors into $Z \in F$ replaces each by a nonconstant binary-valued polynomial $f : F \to \mathbb{F}_2$. Any reduced such polynomial has ordinary degree at least $q/2$. Indeed, if $\deg f < q/2$, then $f^2 - f$ has degree below q and vanishes at every field element, so it is identically zero. Since $F[Z]$ is an integral domain, $f(f-1) = 0$ forces f constant, a contradiction. Trace coordinates attain the degree $q/2$.

Thus the polynomial degree in the field-pencil theorem is an admission condition tested in its source description; it is not preserved by the generic Boolean packing. The universal constant-source, nilpotency-four construction remains universal after packing, but its entries can have exponentially large ordinary degree. This identifies the exact lost cost without declaring another representation impossible.

Allowing an outer trace bypasses the preceding coordinate-value bound for linear functions, but does not remove the general pointwise obstruction. If $\operatorname{Tr} g(Z) = \mathbf{1}_{Z=0}$ on F, then $\deg g \geq q-1$. For smaller degree, field power sums give $\sum_Z g(Z) = 0$, so its trace has even support, contradicting the one-point indicator. The bound is sharp for $\eta(1 + Z^{q-1})$ with $\operatorname{Tr}\eta = 1$. That Boolean indicator has a linear-size conjunction description. More generally, every $\operatorname{Tr} g$ of degree below $q-1$ has an even number of one outputs. A compiler directly preserving the exact count on the same d-bit field domain therefore needs $r \geq 2^d$ whenever the source has an odd accepting count. This excludes that specific direct count-preserving replacement as well as a pointwise one. It does not exclude changed domains, normalization factors, offset readouts, or another counting algorithm. The example indicator's count is elementary.

13 An acquired nonlinear repair for the actual packet interface

The integer packet kernel also admits a concrete nonlinear repair, but its arithmetic-circuit closure is not universally cheap. For multilinear coefficient polynomials A, B and the channel $K = \left(\begin{smallmatrix} 1 & 6035 \\ 0 & -5964 \end{smallmatrix}\right)$, contraction is the Hadamard product of A with $B(6035 - 5964z_1, \ldots, 6035 - 5964z_N)$, evaluated at all ones. If both boundaries are homogeneous products of r linear forms, each variable occurring in one factor on either side, manufacture the interaction matrix W_{ij} by summing the coefficient products of shared wires. Then

$$Z_K = (-5964)^r \operatorname{per}(W).$$

The matched monomials choose one variable per row and column factor. Arbitrary zero-one matrices fit this description. This permanent/Hadamard construction is already established in [6]; a cheap substitution does not make the subsequent Hadamard operation free.

The implemented repair receives the raw renamed factor wires, constructs W, and computes the rank of its *entrywise reciprocal* $R_{ij} = 1/W_{ij}$. All entries must be nonzero. If $\operatorname{rank} R \leq 2$, exact rational linear algebra manufactures scaled Cauchy coordinates

$$W_{ij} = \frac{\rho_i \kappa_j}{x_i + y_j}.$$

For rank two, factor $R_{ij} = u_{i0}v_{j0} + u_{i1}v_{j1}$, and choose an integer $t \in \{0, \dots, 2r\}$ avoiding all zero denominators. The formulas

$$a_i = u_{i0} + tu_{i1}, \quad b_j = v_{j1} - tv_{j0}, \quad x_i = u_{i1}/a_i, \quad y_j = v_{j0}/b_j,$$

with $\rho_i = 1/a_i$, $\kappa_j = 1/b_j$, verify the representation entrywise. Each denominator condition excludes at most one t. Ordinary W can have full, unbounded rank; a low ordinary-rank factorization is not assumed.

Borchardt's classical identity evaluates a nonsingular Cauchy permanent as a quotient of two determinants [7]. Repeated acquired nodes can make this quotient $0/0$. The program constructs the confluent matrices instead: if s_i, t_j are the within-node repetition orders, use

$$C^*_{ij} = \frac{(-1)^{s_i+t_j}\binom{s_i+t_j}{s_i}}{(x_i+y_j)^{s_i+t_j+1}}, \qquad D^*_{ij} = \frac{(-1)^{s_i+t_j}(s_i+t_j+1)\binom{s_i+t_j}{s_i}}{(x_i+y_j)^{s_i+t_j+2}}.$$

Then $\operatorname{per} W = (\prod_i \rho_i \prod_j \kappa_j) \det D^* / \det C^*$, with nonzero denominator. This follows by removing the vanishing row and column Vandermonde factors before taking the coincident-node limits in Borchardt's identity. Rank one is handled directly by $r! \prod \rho_i \prod \kappa_j$. All divisions are exact rational operations before any modular reduction.

The source parsing, acquisition and two fraction-free determinants have polynomial bit cost; with B bounding the rational input entry lengths, $O(r^9(B + \log r)^2)$ is a conservative determinant-stage bound. Deleting a row and a column preserves the gate, so exact edge weights are computed by the same counter. Arbitrarily forbidding edges need not preserve it. No sampler is claimed as executed.

Twenty-one full ordinary-rank cases, seven degeneracies and 46089 directly enumerated packet terms passed. Thirty-six exact marginals matched an independent 720-permutation reference. A dense 32-by-32 example had 1024 original packet coordinates and a 2135-bit exact integer contraction, agreeing with an equally informed classical Borchardt calculation. This is an executed restricted repair with classical ancestry, not a new general permanent counter or a practical speed claim.

14 Manufacturing coefficients erased by trace: quartic norm acquisition

The cubic norm construction has a substantive obstruction at degree four. Over a characteristic-two field, homogeneous quartic squares lose ordinary degree under the absolute trace. For example,

$$\operatorname{Tr}(cX_i^2X_j^2) = \operatorname{Tr}(c^{q/2}X_iX_j), \qquad \operatorname{Tr}(cX_i^4) = \operatorname{Tr}(c^{q/4}X_i) \quad (a \geq 2).$$

Thus fitting the homogeneous quartic alone does not recover its coefficients. Supplying those missing coefficients would assume the useful representation. The following identity manufactures them.

Lemma 14.1 (Quartic Hessian repair)**.** *Let F have characteristic two. Suppose $H(X) = C\prod_{j=1}^4 L_j(X)$ over an algebraic closure, with invertible linear-form coefficient matrix M. Put $\mu = C \det M$, the coefficient of $X_1X_2X_3X_4$. Then*

$$\operatorname{Pf}(\operatorname{Hess} H) = \mu H. \tag{Q1}$$

The Hessian is alternating and is unchanged when all homogeneous square monomials are removed from H.

Proof. For $P(Y) = Y_1Y_2Y_3Y_4$, the three matching terms in the Hessian Pfaffian all equal P, and sum to P in characteristic two. The Hessian transforms as $CM^T(\operatorname{Hess} P)M$. The polynomial identity $\operatorname{Pf}(M^TAM) = \det(M)\operatorname{Pf}(A)$ holds over any commutative ring. Its scalar factor contributes C^2, giving $C^2\det(M)P = \mu H$. First derivatives of polynomial squares vanish. □

Theorem 14.2 (Raw quartic norm compiler)**.** *Input only an explicit Boolean ANF f of degree at most four and its original input count n. For $n = 4a$, $a \geq 5$, the algorithm recognizes and exactly counts every function*

$$f(x) = \operatorname{Tr}_{F/\mathbb{F}_2}\big(CN_{A/F}(\phi(x)) + \operatorname{Tr}_{A/F}(B\phi(x))\big) + \epsilon, \tag{Q2}$$

where $F = \mathbb{F}_{2^a}$, A is any quartic etale F-algebra, $C \neq 0$, $B \in A$, and $\phi : \mathbb{F}_2^n \to A$ is any binary affine bijection. The field, algebra, affine map, coefficients and sign are not inputs. It gives an exact pointwise certificate and counts the original independent inputs. Acceptance requires every raw coefficient to agree.

Construction and proof. The fourth binary polarization is a traced four-dimensional field volume. Solve its full binary centroid equations and acquire the field and a four-element module basis. Nondegeneracy and the centroid test are essential. In these coordinates, trace duality obtains the squarefree quartic coefficient μ from the fourth tensor.

The third binary coefficients uniquely fit 28 field coefficients: twelve monomials $X_i^3X_j$, twelve $X_i^2X_jX_k$, and four squarefree ordinary cubics. The first 24 give the nonsquare quartic other than the known volume term; the last four are translation data. This is binary linear algebra with $28a$ columns, with rank and residual verified. For $a \geq 5$, the relevant Frobenius exponent orbits are distinct. The implementation also verifies the same rank for its smaller accepted fields.

Let H_0 be this manufactured nonsquare quartic. Recover

$$H = \mu^{-1}\operatorname{Pf}(\operatorname{Hess} H_0), \tag{Q3}$$

and require its nonsquare coefficients to equal H_0. This necessary identity is not treated as a sufficient norm certificate.

Test 13 directions $v(s) = (1, s, s^2, s^3)$ at distinct field elements until $H(v) \neq 0$. On a promised norm, each linear factor restricts to a nonzero polynomial of degree at most three, so at most twelve candidates fail. Test 19 bases on the same moment curve until $g(T) = H(u + Tv)/H(v)$ is squarefree. A repeated root equates two ratios $L_i(u)/L_i(v)$. The product of the six pairwise differences restricts to a nonzero polynomial of degree at most 18. Thus the bounded searches succeed with $q \geq 32$.

Manufacture $A' = F[r]/(g(r))$. Its four gradient values $G_i = \partial_i H(u + rv)$ are elements of A'. At each geometric root the gradient is a nonzero scalar times one original linear form, so the map $X \mapsto \sum_i G_iX_i$ is invertible. Verify invertibility and all coefficients of

$$H(X) = C'N_{A'/F}\left(\sum_i G_iX_i\right). \tag{Q4}$$

This manufactures the algebra and proves admissibility without an algebraic-closure computation or a guessed factorization.

If $c_{\widehat{i}}$ is the acquired ordinary cubic coefficient of $\prod_{j\neq i} X_j$, set $t_i = c_{\widehat{i}}/\mu$. Only the squarefree quartic contributes to this coefficient under translation; a differentiated exponent two contributes zero. Translate the raw function, subtract $\operatorname{Tr} H$, and recover every residual Boolean coefficient through degree four. Require an affine residual. The nondegenerate trace pairing on A' acquires its linear element B' and sign. Finally reconstruct the norm expression in the original coordinates and verify all $\sum_{j=0}^{4}\binom{n}{j}$ raw coefficients. This proves correctness for every accepted input. □

For M ANF masks, the source bit length is $L = O(n + nM)$, including the independent-input inventory. There are $O(n^4)$ centroid equations in n^2 binary unknowns. A conservative bound for elimination, field acquisition, coordinate transforms and all coefficient probes is $O(Ln^6 + n^{12})$ bit operations and $O(L + n^8)$ bits of storage. This charges literal scans of the raw ANF. The norm geometry uses a fixed number of constant-degree field operations. The retained one-variable Dwork implementation adds $\widetilde{O}(a^8)$ bit time and $\widetilde{O}(a^4)$ space for counting. These are conservative bounds, not a practical speed comparison.

14.1 Exact quartic count and all etale types

Write $A' = \prod_{i=1}^{s} \mathbb{F}_{q^{d_i}}$, $\sum d_i = 4$, and $B' = (b_i)$. Factor degrees and zero-component degrees are acquired by polynomial gcds with $T^{q^j} - T$ and $B'(T)$; an irreducible factorization is not supplied. For a unit B', the complete signed count is

$$S = (-1)^{\epsilon}(-1)^{4-s} q \left(K_F(N(B')/C')^2 - q\right), \qquad K_F(z) = \sum_{u \in F^*} (-1)^{\mathrm{Tr}(u+z/u)}. \tag{Q5}$$

Davenport–Hasse lifting gives a two-free-variable Kloosterman sum; Carlitz's characteristic-two identity reduces it to $K_F(z)^2 - q$ [5, 8]. These are classical evaluations. The companion proof includes the trivial-character cancellation and the elementary $x + y, xy$ proof of the Carlitz identity.

For a nonunit B', put

$$U_i = \begin{cases} q^{d_i} - 1, & b_i = 0, \\ -1, & b_i \neq 0, \end{cases} \qquad L_0 = \begin{cases} q^4, & B' = 0, \\ 0, & B' \neq 0. \end{cases}$$

Then the complete formula, with an exact integer division, is

$$S = (-1)^{\epsilon} \left(L_0 - \frac{q}{q-1} \prod_i U_i \right). \tag{Q6}$$

These expressions cover all five splitting types and every zero pattern. The Boolean one-count is $(2^n - S)/2$, with no auxiliary-input multiplier.

The backend passed 88 cases across all five types, 1058368 complete original assignments, and nineteen direct Carlitz checks. An explicit 64-bit field norm/linear phase gave $S = -8917155840$ and one-count 9223372041313353728. Its Dwork value $K_{\mathbb{F}_{65536}}(1) = -449$ used 143 matrix states, precision 18 and sixteen Frobenius factors. An independent elliptic Frobenius recurrence agreed. This tests the explicit-field backend; it is not an executed 64-bit raw-acquisition claim.

The full raw compiler passed sixteen positive cases, all five etale types, and 5264640 independently checked original assignments. The 24-bit raw input had 6467 ANF terms and gave $S = -6720$, zero-count 8385248 and one-count 8391968. All 12951 raw coefficients were verified, and an independent F_{64} Kloosterman sum agreed. A quadratic perturbation preserving the fourth tensor was rejected. So was $H = X_0X_1X_2X_3 + X_0^3X_1$, which passes (Q1) but fails the norm-geometry certificate. The finite small-field exceptions use explicitly bounded searches; the uniform $a \geq 5$ construction uses only the 13/19 candidates.

15 A fixed-degree norm reconstruction principle beyond quartics

The repair extends algebraically, with explicit implementation and degree-uniformity limits.

Theorem 15.1 (Even-degree Hessian reconstruction)**.** *Let $k = 2m \geq 4$. Suppose $H = C \prod_{j=1}^{k} L_j$ is a scalar product of k independent linear forms in k variables over an algebraic closure of a characteristic-two field. With $\mu = C \det(L_j)$,*

$$\mathrm{Pf}(\mathrm{Hess}\, H) = \mu H^{m-1}. \tag{E1}$$

Given its nonsquare homogeneous part over a finite field and its nonzero volume coefficient, one manufactures all of H by field arithmetic and linear algebra. For fixed k, the cost is polynomial in field degree. Neither erased coefficients nor a multivariate factorization are supplied.

Proof. For $P = \prod_i Y_i$, each Hessian matching term equals P^{m-1}. The number $(2m-1)!!$ of matchings is odd. Pfaffian covariance gives $C^m \det(M) P^{m-1} = \mu H^{m-1}$.

Let J be the manufactured Pfaffian divided by μ, and write $m - 1 = 2^s t$ with t odd. Recover $K = J^{1/2^s}$ by inverse Frobenius, requiring exponents divisible by 2^s. On a promised input, $K = H^t$. For $t = 1$, this is H. Otherwise solve for a homogeneous degree-k form V by the field-linear equations

$$K\partial_i V = V\partial_i K \quad (1 \le i \le k), \qquad [X_1 \cdots X_k]V = \mu. \tag{E2}$$

Cancel H^{t-1} to obtain $H\partial_i V = V\partial_i H$. For each L_j, choose a coordinate with nonzero coefficient in L_j. Reducing modulo L_j forces $L_j \mid V$, as the other factors are relatively prime to L_j. Thus $H \mid V$. The degrees agree, and the volume coefficient fixes the remaining scalar. Require uniqueness, verify $V^{m-1} = J$ and all supplied nonsquare coefficients. Norm geometry must still be checked. □

There are $B_k = \binom{2k-1}{k}$ degree-k coefficients. The Pfaffian has $(k-1)!!$ matching terms and degree $k(k-2)/2$. System (E2) has B_k unknowns and at most $k\binom{kt+2k-2}{k-1} + 1$ equations. These are constants only for fixed k, not a uniform polynomial bound in k.

Odd-degree Hessian repair. There is a companion identity for $k = 2m + 1 \ge 5$. If p_i is the principal Pfaffian obtained by deleting coordinate i, then

$$p_i(\operatorname{Hess} H) = \mu X_i H^{m-1}. \tag{E3}$$

For the coordinate product, each matching contributes $X_i P^{m-1}$. The principal-Pfaffian vector obeys $p(M^T A M) = \det(M) M^{-1} p(A)$, obtained by bordering the odd alternating matrix and comparing the border-vector coefficients. This proves (E3). In odd degree the Hessian omits precisely the monomials with one odd exponent; they are not polynomial squares. The terms with at least three odd exponents determine the Hessian. Divide each p_i by μX_i, require all quotients to agree, and apply the same perfect-power recovery. Degree five gives H directly and degree seven gives H^2.

The separate odd-degree component passed twelve coefficientwise checks in degrees five and seven. Two dense septics had 1074 and 1126 terms, of which 321 and 356 Hessian-invisible coefficients were manufactured. The general odd-power linear branch is proved and implemented but was not exercised by those degree-five/seven tests. Field acquisition, geometric admission and a complete raw identity remain separate requirements.

Lemma 15.2 (A trace lift without guessed Frobenius coefficients)**.** *Let $q = 2^a > k^2$. Exponent vectors $e \in \mathbb{N}^k$ with $1 \le \sum_i e_i \le k$ and at least one odd coordinate have pairwise disjoint doubling orbits under $X_i^q = X_i$. Each orbit has length a. Consequently $\operatorname{Tr}(cX^e)$, with c over a binary field basis, together with the constant function, are binary linearly independent.*

Proof. Suppose a rotation by 2^r, $0 < r < a$, sends one such vector to another of total degree at most k. If $2^r k < q$, no coordinate wraps and all resulting exponents are even, a contradiction. Otherwise $2^r \ge q/k > k$. An odd coordinate has a bit in position zero which rotates to position r, giving a resulting exponent at least $2^r > k$, another contradiction. Zero coordinates remain zero. This also forbids a shorter orbit. The unique reduced polynomial on F^k has disjoint nonconstant supports for these trace orbits, proving independence. □

Any trace polynomial of ordinary degree at most k enters this canonical form by repeatedly dividing all monomial exponents by two and applying inverse Frobenius to the coefficient. Binary linear algebra acquires the lift from probes of weights at most k. Its top nonsquare part supplies the Hessian in (E1).

The top polarization of a hidden degree-k norm is a traced field volume. For $k \geq 3$ its full centroid is precisely the scalar field. Use three different tensor slots to move a centroid endomorphism and a field scalar past each other. Nondegeneracy forces them to commute, so the endomorphism is field-linear; the centroid of a volume form is scalar. The full binary centroid equations therefore manufacture the field.

The gradient construction (Q4) extends to degree k. A moment-curve direction search needs $k(k-1)+1$ candidates; the base search needs $(k-1)\binom{k}{2}+1$. Take q above these bounds and k^2. After manufacturing H, verify the algebra, its entire norm identity, and the affine remainder. Translation is recovered from the squarefree degree-$(k-1)$ coefficients as before. For fixed k, smaller fields have $q = O(k^3)$, so direct counting costs $q^k = k^{O(k)}$ original assignments. This explicitly charged finite exception is not polynomial in unbounded k. No algebraic closure factorization is assumed.

Counting and its limit. This gives a polynomial raw representation compiler for every fixed even k. In odd fixed degree the homogeneous top part has no square monomials; the canonical lift and geometry test also apply. One may count by the established point counter on $W^2 + W = H(Y) + \ell(Y) + c$, with $k+1$ field variables, degree at most k, and fully manufactured coefficients. Its affine point count is twice the original Boolean zero count. Proposition 36 of [4] gives $\widetilde{O}_k(a^{3k+10})$ bit time and $\widetilde{O}_k(a^{2k+6})$ space, including the fixed 2^{k+1} torus decomposition. A conservative acquisition bound is $O_k(Ln^{k+1} + n^{k+8})$ bit time and $O_k(L + n^{k+4})$ space, with $n = ka$ original independent bits and raw ANF bit length L. Reconstruction and the output integers use $O(n)$ bits.

The general point counter here is a literature algorithm, not an executed implementation. Indeed any acquired bounded-degree polynomial in a fixed number of field coordinates has this generic bound; norm geometry is unnecessary for that theoretical corollary. The norm repair manufactures the algebra and enables the smaller classical backend, implemented here at degree four. It gives neither a uniform polynomial bound in unbounded degree nor a representation compiler for arbitrary circuits.

The executable Hessian/root component passed 22 complete polynomial identities: eighteen dense quartics, three dense sextics and an octic under a controlled invertible shear. The octic used 74587 exact coefficient equations in 6435 unknowns, with full rank. Every canonical exponent was checked for $(k, q) = (4, 32), (6, 64), (8, 128)$, totaling 78670 nontrivial orbit rotations. These are coefficient-level checks, not generic circuit-counting experiments. The data preserve generated inputs and recovery certificates.

16 What remains unresolved and what the files actually execute

The requested polynomial exact counter for arbitrary formulas or circuits has not been constructed. The universal representations in Sections 2 and 10 preserve the answer with polynomial description and exact normalization, but their remaining counting problems have no polynomial bound here. The fully split centroid obstruction in Section 9 is stronger than a generic warning: even that manufactured decomposition retains arbitrary counting in its quadratic glue, with an explicit one-query readout.

The positive operations form reusable exact components: acquired radical restriction, acquired independent blocks, raw etale norm acquisition, and verified unbounded-depth field pencils. They can be composed with their integer factors and original observables intact. Their admission tests are executed, and unsupported terminals are rejected. A new general theorem would still need to

manufacture a polynomially bounded family of admissible terminals, or a different aggregation mechanism for the unrestricted terminal, with all acquisition and reconstruction costs included.

The prior literature already supplies the quadratic/cubic complexity boundary, low-degree phase representations, point-counting engines and special determinant identities [2, 3, 4, 7]. The present proofs and programs make particular acquisition, composition and failure statements concrete. Historical priority and a comparative practical speed advantage are not claimed.

The accompanying source archive contains production programs, the raw inputs, exact output data, independent audits and full companion proofs. It also preserves the previous acquired chain-algebra theorem, its proof PDF and source programs. The previously recovered latest task-directed network files remain unchanged. The preservation note still records that the older exact/noise-aware 510-reaction Gram source was not recovered; this work does not silently fill that provenance gap.

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